

Neural Correlates of Memory for Items and for Associations: An Event-related Functional Magnetic Resonance Imaging Study

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Abstract

■ Although results from cognitive psychology, neuropsychology, and behavioral neuroscience clearly suggest that item and associative information in memory rely on partly different brain regions, little is known concerning the differences and similarities that exist between these two types of information as a function of memory stage (i.e., encoding and retrieval). We used event-related functional magnetic resonance imaging to assess neural correlates of item and associative encoding and retrieval of simple images in 18 healthy subjects. During encoding, subjects memorized items and pairs. During retrieval, subjects made item recognition judgments (old vs. new) and associative recognition judgments (intact vs. rearranged). Relative to baseline, item and associative trials activated bilateral medial temporal and prefrontal regions during both encoding and retrieval. Direct contrasts were then performed between item and associative trials for each memory stage. During en-

coding, greater prefrontal, hippocampal, and parietal activation was observed for associations, but no significant activation was observed for items at the selected threshold. During recognition, greater activation was observed for associative trials in the left dorsolateral prefrontal cortex and superior parietal lobules bilaterally, whereas item recognition trials showed greater activation of bilateral frontal regions, bilateral anterior medial temporal areas, and the right temporo-parietal junction. Post hoc analyses suggested that the anterior medial temporal activation observed during item recognition was driven mainly by new items, confirming a role for this structure in novelty detection. These results suggest that although some structures such as the medial temporal and prefrontal cortex play a general role in memory, the pattern of activation in these regions can be modulated by the type of information (items or associations) interacting with memory stages. ■

INTRODUCTION

An increasing amount of empirical evidence suggests that episodic memories are formed of two main types of information: (1) information about individual items and (2) information about the associations that these items maintain with each other and with their physical (e.g., color) or spatio-temporal (e.g., position in space) characteristics. These two types of information have been termed *item memory* and *associative (relational) memory*, respectively (Murdock, 1982; Humphreys, 1976, 1978). Although item and associative memory obviously share some common underlying episodic memory processes, there could also be some difference in the processing of these two types of information. For instance, the longer response times observed during the retrieval of associative information relative to item information (Hockley, 1991; Gronlund & Ratcliff, 1989), the greater persistence of associative memory over time (Hockley, 1991, 1992), and the documentation of distinct receiver operating characteristics (Yonelinas, 1997) attests to the

view that item and associative information are distinctively processed in memory. It follows that memory for these two types of information could rely, at least in part, on different brain regions.

Already, there are a number of findings that support this inference. Lesion studies in both rodents (Brown & Aggleton, 2001; Eichenbaum, Otto, & Cohen, 1994; Cohen & Eichenbaum, 1993) and humans (Mayes et al., 2001; Mayes, van Eijk, Gooding, Isaac, & Holdstock, 1999; Vargha-Khadem et al., 1997) suggest that the hippocampus plays a critical role in associative memory, but not in item memory. However, item memory is not completely spared in patients with hippocampal damage, and some have argued that the hippocampus could be similarly involved in both types of memory (Stark, Bayley, & Squire, 2002; Stark & Squire, 2000, 2001). The prefrontal cortex could also be involved in associative memory as suggested by other neuropsychological studies that have shown that patients with frontal lobe lesions are relatively more impaired in tasks of associative memory relative to item memory (Johnson, O'Connor, & Cantor, 1997; Janowsky, Shimamura, & Squire, 1989). Lesion studies, however, do not give information about which memory stage is affected. Indeed, the deficits could

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reflect that (1) associative information is not successfully encoded and, thus, cannot be retrieved, (2) associative information has been correctly encoded but cannot be accessed because of retrieval deficits, or (3) both associative encoding and retrieval are impaired. Thus, studying the distinction between item and associative memory during both encoding and retrieval should lead to a better understanding of the processes that support these two types of memory. More specifically, some brain regions could be involved in item or associative memory regardless of the memory stage, but other brain regions could be specifically involved in encoding or retrieval of a certain type of information.

An important advantage of functional brain imaging techniques is that they enable us to examine encoding and retrieval separately while looking at the whole brain. During encoding, two main strategies have been adopted to distinguish brain activation specific to associative or item-oriented encoding. The first strategy is to give the subjects different instructions, with one set of instructions promoting associative encoding and the other set of instructions promoting individual or nonassociative encoding. Another strategy is to present stimuli that either encourage (e.g., pairs of items) or hinder (e.g., individual items) associative encoding. The neuroimaging studies that have used either of these two strategies have consistently reported greater hippocampal and prefrontal activation for associative relative to item encoding (Davachi & Wagner, 2002; Davachi, Maril, & Wagner, 2001; Killgore, Casasanto, Yurgelun-Todd, Maldjian, & Detre, 2000; Henke, Weber, Kneifel, Wieser, & Buck, 1999; Montaldi et al., 1998; Henke, Buck, Weber, & Wieser, 1997; Kapur et al., 1996; Vandenberghe, Price, Wise, Josephs, & Frackowiak, 1996).

During recognition, the most consistent finding is greater left prefrontal activation for associative relative to item recognition (Lepage, Brodeur, & Bourgouin, 2003; Badgaiyan, Schacter, & Alpert, 2002; Rugg, Fletcher, Chua, & Dolan, 1999; Nolde, Johnson, & D'Esposito, 1998; Cabeza et al., 1997), although right prefrontal activation has also been observed (Lepage et al., 2003; Rugg et al., 1999; Cabeza et al., 1997). In contrast to the consistent prefrontal activation, only a few studies have reported medial temporal activation for associative recognition relative to item recognition (Tsukiura et al., 2002; Yonelinas, Hopfinger, Buonocore, Kroll, & Baynes, 2001). These studies did not, however, use a typical associative recognition procedure in which subjects are presented with both intact and rearranged associations and required to distinguish between these two types of associations. To illustrate this typical procedure further, if the subjects studied the associations A–B, C–D, and E–F, then they should categorize A–B as intact but C–F as rearranged. These intact and rearranged associations can be formed of 2 (or more) items (memory for pairs) or of an item and an associated feature such as space or color (source memory). In the previous example, if A is an

item, B could be either another item or a feature associated with A.

In the study by Yonelinas et al. (2001), individual items (black and white images) were presented during the recognition test and subjects were asked to judge whether each item was old or new (item recognition) or to judge whether each stimulus was initially presented in red or green (associative recognition). The associative recognition task did not require subjects to distinguish between intact and rearranged associations but instead it required them to recall the color initially associated with each item. When these associative recognition trials were contrasted to old item recognition trials, hippocampal activation was observed. It is, therefore, possible that the demands of generating the initial association, not the recognition of associative information per se, resulted in hippocampal activation in this experiment. In the study by Tsukiura et al. (2002), the associative recognition task that gave rise to left parahippocampal activation might have also involved the generation of additional associative information. During encoding, the subjects were presented with comic strips formed of 4 images. During recognition, they were cued with 2 images and asked to judge whether these images were from the same strip (associative recognition) or whether both images had been studied before (item recognition). It is quite possible that the subjects tried to remember the missing images from the same strip to facilitate their associative recognition judgments. Thus, although the hippocampus is clearly implicated in associative encoding, it is still unclear whether activity in the medial temporal lobe can be justifiably linked to associative recognition.

Because episodic memory encoding and retrieval are supported by partially different brain regions, and because the activation of specific brain regions (e.g., those implicated in the processing of item or associative information) typically depends on the pattern of activation elsewhere in the brain (McIntosh, 2000), brain regions supporting item and associative memory are likely to differ between encoding and retrieval. It follows that studying both encoding and retrieval in the same experiment should lead to a better understanding of item and associative memory. Moreover, most neuroimaging studies of memory encoding and recognition have placed an emphasis on associative memory, and the brain regions contributing specifically to item relative to associative memory are not consistently reported (e.g., Lepage et al., 2003; Yonelinas et al., 2001; Henke et al., 1999; Vandenberghe et al., 1996). Preferential activation for items could nonetheless potentially be observed during encoding, retrieval, or both memory stages because of the nature of the stimulus that the person is faced with (i.e., items) and/or the distinct type of processing that the subject is engaged in (i.e., the task orientation; Rugg & Wilding, 2000) when cued by the appearance of this type of stimulus. Our understanding

of episodic memory could, thus, be further clarified by looking at the brain activations specific to both types of information, particularly if this is done during both encoding and retrieval stages.

The present study used event-related functional magnetic resonance imaging (fMRI) to assess the neural correlates of both item and associative memory during both encoding and retrieval in 18 healthy participants. During the encoding phase, participants were presented with items (images duplicated to have the same layout as the pairs, item encoding) and pairs (2 different images side by side, associative encoding). They were instructed to report whether an item or a pair was presented and to memorize the images (for both items and pairs) and their associations (for the pairs only). During the retrieval phase, participants were again presented with items and pairs and were required to make memory judgments as follows: (1) when items were presented (item recognition), subjects were required to indicate with a mouse click whether it was old (studied before) or new (never studied before), and (2) when pairs were presented (associative recognition), subjects were instructed to indicate with a mouse click whether it was intact (images presented in the same pairing as in the encoding session) or rearranged (images from previously studied pairs presented in a rearranged pairing). This design allowed us to assess, using event-related fMRI, brain regions distinguishing between the 2 types of information while using the same type of stimuli during both encoding and retrieval. A pair of abstract images was randomly presented during both encoding and recognition and was used as a baseline.

Contrasts were first performed between the memory trials (items and associations) and the baseline during each memory stage to assess the brain regions involved in memory for these 2 types of information. The effect of type of information (item vs. associative) was then considered separately for each memory stage, revealing brain regions that responded to items and associations with different patterns of activation during encoding or retrieval but not necessarily both. Conjunction analyses were performed to detect regions of common activation for both the encoding and recognition of each type of information relative to the other type. The combined probability in the conjunction analyses led to lower critical t values associated with the same level of significance (e.g., a voxel significant at 0.1 in 2 occasions has a probability of 0.01 for their conjunction), thus enabling some regions not significantly activated in encoding or recognition contrasts to reach the significance level in the conjunction analyses. It is, thus, important to keep this difference in mind when interpreting the results from the contrasts and the conjunction analyses.

For the comparisons against the baseline, we hypothesized that activity in brain regions implicated in episodic memory such as the medial temporal lobe and prefrontal cortex during both encoding and recognition

would be observed. Our main interest resided in the comparison between item and associative trials. We expected to observe a modulating effect of the type of information (item vs. associative) during both encoding and retrieval. Based on previous neuroimaging experiments, we anticipated preferential activation of the hippocampus and prefrontal cortex for associative relative to item encoding. At recognition, we predicted that associative information would activate the prefrontal cortex, most likely in the left hemisphere. Hippocampal activation for associative relative to item recognition would be expected if this region indeed contributes to processes implicated in the recognition of associations. However, as mentioned earlier, we believe that the role of the hippocampus at this stage could be related to the generation of associated information that is not supplied by the recognition test, a process that is not explicitly called for by our associative recognition test. Along with the activation specific to associative encoding and retrieval, we also expected to observe regions of greater activation for item information. Even if the associations are also composed of items, similar stimuli can be processed differently when they are presented in an item or an associative memory task. During retrieval, for instance, assessment of novelty/familiarity can be used as a basis for item recognition judgments but not for associative recognition judgments (Yonelinas, 2002). It has been proposed that item memory could rely on the parahippocampal gyrus (Eichenbaum et al., 1994), but we believe that other brain regions, and in particular some prefrontal areas, could also support primarily item encoding and/or recognition.

RESULTS

Behavioral Results

Performance on the encoding task was almost perfect, with 99% ($SD = 1\%$) of the items and pairs being correctly classified. The overall recognition accuracy, as defined by the mean hit plus correct rejection performance, was 80% ($SD = 10\%$), with subjects being 87% ($SD = 7\%$) accurate on the item recognition task and 72% ($SD = 15\%$) accurate on the associative recognition task. A t test revealed a significant mean difference between item and associative recognition, $t(17) = 6.21$, $p < .0001$.

The mean response time for the encoding task was 1214 msec ($SD = 254$). For the recognition task, the mean response time was of 1492 msec ($SD = 151$), with 1324 msec ($SD = 156$) for the item recognition task and 1664 msec ($SD = 172$) for the associative recognition task. A t test revealed a significant difference between item and associative recognition, $t(17) = 11.18$, $p < .0001$. The response times differed between intact (1598 msec, $SD = 193$) and rearranged (1731 msec, $SD = 170$) pairs, $t(17) = -4.74$, $p < .0001$, but not between old

(1292 msec, $SD = 170$) and new (1354 msec, $SD = 168$) items, $t(17) = -1.98$, $p = .064$, bidirectional.

Functional Magnetic Resonance Imaging Results

For the encoding contrasts, all trials were included in the analyses (items = 1080, associations = 1080, and baseline = 1080). For the recognition contrasts, only the trials associated with correct responses were included in the analyses (items = 940, associations = 778, and baseline = 1080).

Memory trials (items and associations) were first contrasted with the baseline of their respective runs during both encoding and recognition. The brain regions activated for the encoding trials are presented in Table 1 and included brain areas generally implicated in visual memory encoding, such as the medial temporal lobe and prefrontal cortex bilaterally, but also bilateral visual areas, the left motor cortex, and the thalamus bilaterally.

The brain regions activated during correct recognition trials relative to baseline are presented in Table 2. These regions included bilateral medial temporal and prefrontal

Table 1. Activation Elicited by the Encoding of Both Items and Associations Relative to Baseline

Peak <i>t</i> Value	<i>x</i>	<i>y</i>	<i>z</i>	Hemisphere	Region	BA
6.13	-38	24	20	Left	Middle frontal gyrus	9
7.19	-42	-4	12	Left	Insula	13
4.37	-28	-6	-6	Left	Putamen	
2.46	-30	-12	-28	Left	Hippocampus	
2.52	-30	-14	-26	Left	Parahippocampal gyrus	
2.97	-12	-18	-20	Left	Hippocampus	
7.83	-10	-20	10	Left	Thalamus	
4.45	-10	-24	46	Left	Posterior cingulate	31
10.53	-36	-28	56	Left	Precentral gyrus	4
5.02	-18	-30	-2	Left	Thalamus	
12.26	-28	-64	-14	Left	Parahippocampal/fusiform gyrus	19, 37
3.50	34	38	36	Right	Middle frontal gyrus	9
5.67	46	30	26	Right	Middle frontal gyrus	46
5.30	34	24	4	Right	Insula	13
5.99	54	6	30	Right	Inferior frontal gyrus	9
8.28	2	2	52	Right	Medial frontal gyrus/cingulate gyrus	6, 32
4.50	8	2	4	Right	Caudate	
4.87	52	0	42	Right	Middle frontal gyrus	6
5.64	42	-2	14	Right	Insula	13
3.25	26	-2	-22	Right	Amygdala	
4.43	20	-4	-2	Right	Globus pallidus	
3.80	42	-10	-30	Right	Hippocampus/parahippocampal gyrus	
5.72	34	-14	64	Right	Precentral gyrus	6
6.40	10	-16	10	Right	Thalamus	
2.65	20	-16	-20	Right	Hippocampus	
4.53	62	-30	26	Right	Inferior parietal lobule	40
10.18	22	-50	-20	Right	Parahippocampal/fusiform gyrus	19, 37

BA = Brodmann Area.

x, *y*, and *z* coordinates of local maxima are listed according to the Talairach coordinate system.

Table 2. Activation Elicited by Both Item Recognition and Associative Recognition Relative to Baseline

<i>Peak t Value</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>Hemisphere</i>	<i>Region</i>	<i>BA</i>
3.46	−22	34	−16	Left	Orbito-frontal gyrus	11
7.91	−40	22	24	Left	Middle frontal gyrus	46
5.41	−10	4	0	Left	Globus pallidus	
7.88	−2	−4	56	Left	Medial frontal gyrus	6
4.48	−22	−12	4	Left	Globus pallidus	
2.40	−24	−12	−22	Left	Hippocampus	
4.24	−36	−18	−24	Left	Hippocampus/parahippocampal gyrus	
6.81	−12	−20	10	Left	Caudate	
6.08	−28	−20	72	Left	Precentral gyrus	4
4.54	−30	−24	6	Left	Temporo-parietal junction	
4.16	−32	−24	−22	Left	Hippocampus	
8.81	−46	−34	46	Left	Inferior parietal lobule	40
9.97	−32	−48	−20	Left	Parahippocampal/fusiform/occipital	18/19/37
7.61	−32	−52	48	Left	Superior parietal lobule	7
3.65	32	62	0	Right	Middle frontal gyrus	10
4.24	30	58	−12	Right	Orbito-frontal gyrus	11
3.23	36	52	12	Right	Middle frontal gyrus	10
7.76	32	22	2	Right	Insula	13
6.53	12	4	2	Right	Globus pallidus	
2.38	28	−2	−12	Right	Amygdala	
5.82	38	−4	60	Right	Middle frontal gyrus	6
4.31	38	−4	10	Right	Insula	13
4.35	22	−28	0	Right	Thalamus	
5.45	46	−34	48	Right	Inferior parietal lobule	40
10.67	28	−56	−12	Right	Parahippocampal/fusiform/occipital/cerebellum	18/19/37

BA = Brodmann Area.

x, *y*, and *z* coordinates of local maxima are listed according to the Talairach coordinate system.

tal areas generally implicated in recognition memory. Activation was also observed in bilateral visual areas, bilateral parietal areas, and the left motor cortex, as well as in some subcortical structures.

Contrasts between item and associative memory trials were then examined separately for encoding and recognition. Results of the comparison between item and associative encoding trials are presented in Table 3 and Figure 1A. Relative to item encoding, associative encoding yielded significant activation of the right hippocampus, right prefrontal cortex, right anterior and posterior cingulate gyrus, right precuneus, and left middle frontal gyrus. No region of significant activation was detected for item encoding relative to associative encoding at the selected statistical threshold.

Results of the comparison between correct item and correct associative recognition trials are presented in Table 4a and Figure 1B,C. Relative to item recognition, associative recognition resulted in activation in the left middle frontal gyrus and bilateral superior parietal lobule, but no significant hippocampal activation was detected for this contrast. When item recognition was contrasted to associative recognition, however, activation was observed in the right amygdala, bilateral anterior hippocampi, right parahippocampal gyrus, as well as bilateral anterior superior frontal gyri, right orbito-frontal gyrus, right middle frontal gyrus, bilateral anterior cingulate, and right temporo-parietal junction. When only old items and intact pairs were included in the analyses, the medial temporal activation observed for

Table 3. Effect of the Type of Information (Associative vs. Item) on the Pattern of Brain Activation at Encoding

Peak <i>t</i> Value	<i>x</i>	<i>y</i>	<i>z</i>	Hemisphere	Region	BA
<i>Associative encoding > Item encoding</i>						
3.66	-24	36	52	Left	Middle frontal gyrus	8
6.44	24	58	-10	Right	Superior frontal gyrus	10
4.12	26	36	36	Right	Middle frontal gyrus	9
3.67	30	30	-4	Right	Inferior frontal gyrus	47
4.73	6	8	40	Right	Anterior cingulate	24
4.53	6	2	54	Right	Medial frontal gyrus	6
5.16	14	-24	44	Right	Posterior cingulate	31
2.41	28	-26	-16	Right	Hippocampus	
4.34	2	-58	42	Right	Precuneus	7
<i>Item encoding > Associative encoding</i>						
No region of significant activation was detected at the selected thresholds						

BA = Brodmann Area.

x, *y*, and *z* coordinates of local maxima are listed according to the Talairach coordinate system.

item recognition disappeared, and no medial temporal activation was observed for associative trials. Furthermore, the comparison of new relative to old item trials revealed bilateral anterior medial temporal activation, confirming that new items elicited greater activation in this region. Results from these post hoc analyses of the medial temporal area are presented in Table 4b.

Conjunction analyses were then performed to assess the brain regions specifically involved in item or associative memory regardless of the memory stage. The regions of significant activation are presented in Table 5. For associative relative to item information, regions of common activation between encoding and retrieval were observed in the anterior cingulate, right inferior

Figure 1. Pattern of activity for (A) associations relative to items during encoding, (B) associations relative to items during recognition, and (C) items relative to associations during recognition. The left and right columns present the activation maps superimposed onto the left and right hemispheres of a canonical brain, respectively. The central column presents the patterns of medial temporal activation superimposed onto coronal slices of a canonical brain. For these coronal slices, *y* values denote the coordinates in the Talairach coordinate system.

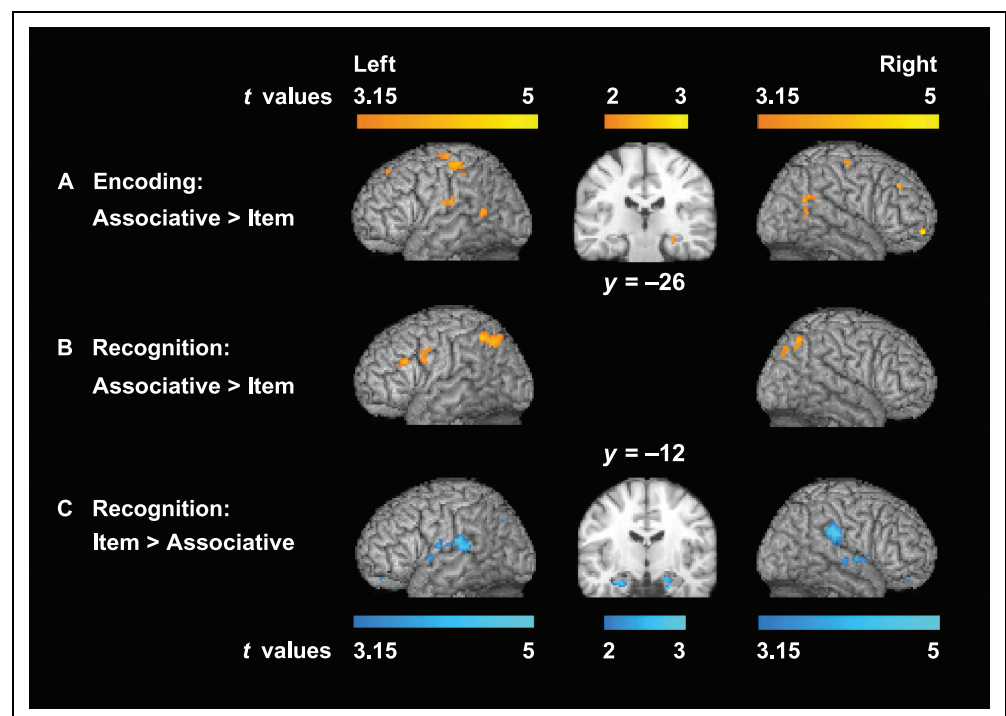


Table 4. Effect of the Type of Information (Associative vs. Item) on the Pattern of Brain Activation at Recognition and Post Hoc Analyses in the Medial Temporal Region

<i>Peak t Value</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>Hemisphere</i>	<i>Region</i>	<i>BA</i>
<i>(a) Effect of the Type of Information (Associative vs. Item) on the Pattern of Brain Activation at Recognition</i>						
<i>Associative recognition > Item recognition</i>						
3.97	−44	22	26	Left	Middle frontal gyrus	46
4.43	−32	−54	48	Left	Superior parietal lobule	7
4.46	28	−62	46	Right	Superior parietal lobule	7
<i>Item recognition > Associative recognition</i>						
3.94	−8	56	16	Left	Medial frontal gyrus	10
4.29	−22	12	32	Left	Anterior cingulate	32
3.80	−28	−10	−22	Left	Hippocampus	
3.35	16	48	24	Right	Superior frontal gyrus	9
3.90	4	44	−18	Right	Orbito-frontal gyrus	11
4.06	32	20	38	Right	Precentral gyrus	8
5.00	16	2	30	Right	Anterior cingulate	24
2.96	28	−4	−24	Right	Amygdala	
2.90	20	−14	−30	Right	Hippocampus/parahippocampal gyrus	
5.01	56	−28	26	Right	Temporo-parietal junction	40/22
<i>(b) Post Hoc Analyses in the Medial Temporal Region ($p < .01$)</i>						
<i>Old item recognition > Intact pair recognition</i>						
No region of significant activation was detected in the medial temporal lobe at the selected threshold						
<i>New item recognition > Old item recognition</i>						
2.57	−26	−8	−8	Left	Amygdala	
2.50	−36	−24	−26	Left	Parahippocampal gyrus	
3.00	32	−2	−16	Right	Amygdala	

BA = Brodmann Area.

x, *y*, and *z* coordinates of local maxima are listed according to the Talairach coordinate system.

frontal gyrus, and left precentral gyrus. For item relative to associative information, activation was observed in bilateral amygdala and anterior hippocampi.

DISCUSSION

This fMRI study aimed to identify brain regions that are differentially activated when items or associations are processed in episodic memory. Neural correlates of item and associative memory were first assessed separately for encoding and retrieval. Then, a conjunction analysis revealed brain regions preferentially activated for each type of information regardless of the memory

stage. One of the main findings is that the type of information that preferentially recruits the medial temporal lobe is not the same during encoding and retrieval. A region in the body of the right hippocampus was activated during associative encoding, but bilateral anterior medial temporal activation (amygdala and anterior hippocampus) was observed for item recognition as well as in the conjunction analysis. Another central finding is related to the pattern of prefrontal activation. During encoding, greater bilateral prefrontal activation was observed for associative information, whereas at retrieval, the laterality of prefrontal activation was related to the type of information. It also worth mentioning

Table 5. Conjunction of Encoding and Recognition of Each Type of Information (Associative vs. Item) Relative to the Other Type

<i>Peak t Value</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>Hemisphere</i>	<i>Region</i>	<i>BA</i>
<i>(Associative > Item at encoding) ∩ (Associative > Item at recognition)</i>						
3.34	−6	12	46	Left	Anterior cingulate	32
2.83	−28	−10	56	Left	Precentral gyrus	6
2.53	30	24	−2	Right	Inferior frontal gyrus	47
<i>(Item > Associative at encoding) ∩ (Item > Associative at recognition)</i>						
1.88	−16	−10	−18	Left	Amygdala	
1.65	−32	−14	−28	Left	Hippocampus/parahippocampal gyrus	
1.92	22	−8	−16	Right	Amygdala	
2.00	20	−10	−26	Right	Hippocampus	

BA = Brodmann Area.

x, *y*, and *z* coordinates of local maxima are listed according to the Talairach coordinate system.

that parietal activation was observed for associative information during both encoding and retrieval, but distributed more medially at encoding and more laterally at retrieval. Differences between item and associative memory were, thus, observed during both encoding and retrieval, but the brain regions distinguishing between item and associative information generally differed between stages. Overall, these results clearly demonstrate the importance of taking the memory stage into account when neural correlates of item and associative memory are considered.

Item and Associative Encoding

During the encoding stage, right hippocampal, bilateral prefrontal, and medial parietal activation was observed for associative information relative to item information, but no region showed significantly more activation for item than for associative encoding at the selected threshold. The lack of significant activation for item encoding relative to associative encoding could reflect the fact that the items composing the associations were encoded along with associative information, thus canceling out the activation specific to item encoding. Our participants were indeed instructed to remember both the items and their associations when pairs were presented. It, thus, remains a challenge for future studies to better isolate the neural correlates specific to item encoding.

In the prefrontal cortex, different regions located mainly in the right hemisphere showed greater activation for associative relative to item trials. Although prefrontal activation has been reported in several neuroimaging studies of memory encoding, right activation has been reported mostly for nonverbal stimuli (Kelley, Buckner, & Petersen, 1998). In our study,

images depicting common objects and animal were used for both associative and item stimuli. It is, thus, possible that the encoding strategy used for associative encoding involved greater mental manipulation of the images to integrate them into a unified representation. Indeed, preferential activation for associations relative to items was observed in the medial parietal cortex, a region that has been suggested to play a role in mental imagery (Wheeler, Petersen, & Buckner, 2000; Fletcher, Shallice, Frith, Frackowiak, & Dolan, 1996; Mellet et al., 1996).

For associative encoding, hippocampal activation is consistent with previous neuroimaging studies of associative relative to item encoding (Davachi & Wagner, 2002; Davachi et al., 2001; Henke et al., 1999; Henke et al., 1997; Vandenberghe et al., 1996) and likely reflects the creation of an association between different pieces of information. Hippocampal activation at encoding has also been shown to predict subsequent associative recognition success (Jackson & Schacter, 2004), supporting the implication of this structure in the creation of durable associations in memory.

Two potential confounds should, however, be addressed. The first potential confound is the deeper or more elaborate processing performed on associations relative to items at encoding. It should be noted though that some previous studies failed to reveal hippocampal activation for deep as compared with shallow encoding when the deep condition was not associative (Petersen, Sandblom, Elfgrén, & Ingvar, 2003; Wagner et al., 1998; Kapur et al., 1994), suggesting that the creation of associations in memory, and not simply the deeper processing at encoding, relies on the hippocampus. This idea would nonetheless have to be directly tested. Another confound is the presence of 2 different images in our associative condition relative to 2 similar images

in our item condition. The activation could, thus, reflect encoding of additional information, regardless of the associative component. The pattern of activation observed in this study is nonetheless consistent with that of other studies that employed the same type of stimuli for associative and item encoding (but different sets of instruction).

Item and Associative Recognition

During recognition, left frontal and bilateral parietal activation was observed for associative information. On the other hand, bilateral frontal, bilateral anterior medial temporal, as well as the right temporo-parietal junction showed greater activation for item information. It has been proposed that associative recognition judgments rely primarily on conscious recollection, whereas item recognition judgments can rely on both conscious recollection and familiarity assessment (Yonelinas, 1997, 2002; Humphreys, 1978). It is also thought that conscious recollection relies on the hippocampus and prefrontal cortex, whereas familiarity assessment relies on the regions surrounding the hippocampus (Yonelinas, 2002). The present results suggest that at least one of these two ideas may need to be revisited because we observed greater hippocampal activation for item recognition relative to associative recognition. Two neuroimaging studies have implicated the hippocampus in conscious recollection (Yonelinas et al., 2001; Eldridge, Knowlton, Furmanski, Bookheimer, & Engel, 2000). In both studies, however, the condition involving greater conscious recollection also involved generating information that was not presented in the recognition test. In the study by Yonelinas et al. (2001), subjects were presented with a black-and-white image and asked to remember the color (red or green) in which that item was originally presented. In the study by Eldridge et al. (2000), subjects were asked to classify old items as “remembered” if they could recollect perceptual details they noticed or associations they made with the word during the study phase. As such, some information had to be generated for an old item to be classified as remembered. Similarly, hippocampal activation has been reported in studies of cued recall where subjects are presented with a cue and are asked to generate the associated information (e.g., Henke, Mandadori, et al., 2003; Henke, Treyer, et al., 2003; Schacter, Alpert, Savage, Rauch, & Albert, 1996). Hippocampal activation was even observed in two different studies where the associations had been learned without awareness (i.e., in a masking procedure) and the subjects were asked to guess the associated information (Henke, Mandadori, et al., 2003; Henke, Treyer, et al., 2003). It, thus, seems that conscious recollection is not required to produce hippocampal activation during associative retrieval. If hippocampal activation was instead associated with the generation of associated information, it could explain

why the hippocampus is not more active for associative relative to item recognition in this as well as some other studies (e.g., Henson, Rugg, Shallice, Josephs, & Dolan, 1999) where both members of the association are presented as a cue at retrieval.

The low success rate in our associative recognition condition probably led to the inclusion in our analyses of correct guesses that were not supported by conscious recollection. It has been previously shown that performance can significantly affect the pattern of brain activation (Callicott et al., 2003). This could have contributed to the lack of hippocampal activation for associative recognition, but does not provide a sufficient explanation for the greater hippocampal activation observed for item recognition. A possible explanation for the anterior hippocampal and amygdala activation observed for item recognition is that it reflects novelty detection and further encoding that occurs at the time of retrieval. Novelty detection has indeed been repeatedly reported to activate the anterior medial temporal lobe (Henson, Cansino, Herron, Robb, & Rugg, 2003; Kohler, Crane, & Milner, 2002; Strange & Dolan, 1999; Strange, Fletcher, Henson, Friston, & Dolan, 1999). At the time of retrieval, novelty detection and further encoding would mainly occur for the new items presented as lures (Habib & Lepage, 2000). Supporting this idea, hippocampal activation was no longer observed when only old items and intact associations were included in our analyses, and a direct contrast between new and old items revealed significant activation for new items in the anterior medial temporal region. Novelty detection can be regarded as an index of familiarity, which can be used as a basis for making item recognition judgments but not associative recognition judgments. It is, thus, not surprising to observe hippocampal activation related to novelty detection during item recognition.

Still, it is quite possible that different regions of the hippocampal formation would support different processes at the time of retrieval. The bilateral hippocampal activation observed for item recognition in this study was located in the anterior part of the structure (peaks at $y = -14$). It is, thus, possible that activation in a more posterior region of the hippocampal formation could support associative retrieval processes, such as the generation of associated information.

In the prefrontal cortex, left dorsolateral prefrontal activation has been previously observed in several studies contrasting associative to item recognition (Lepage et al., 2003; Badgaiyan et al., 2002; Rugg et al., 1999; Nolde, Johnson, & D’Esposito, 1998) and could support retrieval of the link between the components of the associations. It has been proposed that the left prefrontal cortex is required for recognition tasks that require processing of significant relational information, whereas the right prefrontal cortex would be more likely implicated in tasks that involve little or no relational process-

ing but instead involve recognizing details or individual components of previously experienced events (Badgaiyan et al., 2002; Tsukiura et al., 2002). A similar model has been proposed by Johnson and Nolde's group (Raye, Johnson, Mitchell, Nolde, & D'Esposito, 2000; Nolde, Johnson, & D'Esposito, 1998; Nolde, Johnson, & Raye, 1998), who distinguished two types of processes that occur at the time of retrieval: (1) heuristic processes, which are relatively simple and quick and are sufficient to perform easy recognition tasks, and (2) systematic processes engaged when the task demand increases or when more detailed evaluation is required at the time of retrieval, as is the case in associative recognition. According to their model, heuristic processes would rely mainly on the right prefrontal cortex whereas systematic processes would also engage the left prefrontal cortex. The present results are consistent with these 2 models, although a region of the left anterior prefrontal cortex was also active for item retrieval.

Bilateral superior parietal activation was also observed for associative recognition trials. Parietal activation has been observed in several neuroimaging studies of episodic memory retrieval and is thought to be an index of retrieval success because it is consistently observed for "old" relative to "new" recognition trials (Rugg & Henson, 2002). In our study, the images forming both intact and rearranged pairs had been studied during the encoding phase, although only the old items had been studied before. It is, thus, possible that the greater familiarity with the images presented during the associative recognition test led to this pattern of parietal activation. An alternative explanation is that parietal activation could reflect the greater task demands of associative recognition trials. Parietal activation has indeed been shown to be positively correlated with reaction time during a working memory task (Honey, Bullmore, & Sharma, 2000). It is possible that the longer response times for associative recognition trials, or the greater difficulty that led to longer response times, elicited superior parietal activation. This idea is congruous with the suggestion that parietal activation participates in response selection by maintaining a representation of competing responses while the decision is being made (Bunge, Hazeltine, Scanlon, Rosen, & Gabrieli, 2002).

Conjunction Analyses

The brain regions recruited by each type of information were not the same during encoding and retrieval, and only a few regions showed significant activation in the conjunction analysis, even with the lower *t* value threshold associated with the combined probability. Nonetheless, greater activation of the amygdala and anterior hippocampus for items relative to associations in the conjunction analysis is noteworthy because the hippocampus is usually thought to be more active for

associations than for items regardless of the memory stage. Furthermore, the frontal activation observed for associative memory in the conjunction analysis suggests that this region could be involved in detecting and processing associative information regardless of the memory stage.

Episodic Memory for Item and Associative Information

By looking at both memory stages together as well as separately, the present study sheds new light on our understanding of item and associative memory. As already mentioned, this study demonstrates the importance of taking the memory stage into account when trying to explain the differences in memory for these two types of information. Obviously, memory for items and for associations share common underlying memory processes, and some brain structures were recruited for both associative and item memory relative to our nonmemory baseline. At the same time, some processes seem to be recruited specifically when associations or items are processed in memory. It has already been proposed that the hippocampus is more important for the acquisition of new episodic memories than for the retrieval of past memories (Simons & Spiers, 2003). During episodic memory encoding, items or associations are processed together with their context. At this stage, the hippocampus would act as a detector of novelty and would link stimuli and context together to form an episodic memory. When associations are presented or when the emphasis is on relational encoding, more links would be formed and the hippocampus would be especially recruited. Prefrontal activation at this stage seems to be tied to the encoding strategy, with additional activation for deeper processing.

During recognition, bilateral prefrontal regions would support memory processes, with left prefrontal activity increasing as the task demands increase (Tsukiura et al., 2002; Raye et al., 2000; Nolde, Johnson, & D'Esposito, 1998; Nolde, Johnson, & Raye, 1998). The left prefrontal region was activated in our associative recognition condition and is consistently reported as being active in studies contrasting associative to item recognition (Rugg et al., 1999; Nolde, Johnson, & D'Esposito, 1998) or contrasting remember to know judgments (Eldridge et al., 2000; Henson, Rugg, et al., 1999).

We found no evidence of a greater involvement of the hippocampus in associative recognition as compared with item recognition. Instead, item recognition activated the hippocampus and surrounding medial temporal cortex bilaterally, presumably reflecting novelty detection and the further encoding of these new stimuli.

The definition of associative memory varies between research groups (e.g., Giovanello, Schnyer, & Verfaellie, 2004; Lepage et al., 2003; Nolde, Johnson, & D'Esposito, 1998; Yonelinas, 1997), and it is not clear whether dif-

ferent types of associations all rely on the same brain regions. It is therefore possible that interitem associations are processed differently than associations between an item and an associated feature such as location or color. The hippocampus could, thus, be implicated in recognition of the association between an item and its spatio-temporal context, but not in the recognition of pairs of items. A recent study by Düzel et al. (2003) has indeed revealed right hippocampal activation when contrasting the recognition of item–space associations (i.e., the association between an item and its position in space) to the recognition of interitem associations (i.e., associations between items).

It has been suggested that when the elements forming the association can be fused together or interpreted as a unified structure, then the hippocampus might not be required to perform associative memory tests (Eichenbaum & Bunsey, 1995; Eichenbaum et al., 1994). It is possible that interitem associations are more subject to fusion than some other types of associations, a view that may explain the lack of hippocampal activation during associative recognition in our study. It would be interesting in future experiments to compare the pattern of brain activation for different types of associations during both encoding and retrieval.

Another variable that could influence the pattern of brain activation is whether each element is used to form only one association or takes part in many associations. For example, in the study by Yonelinas et al. (2001), each of the two colors (red and green) were associated with many items. In other studies, elements did not overlap between the associations (i.e., they took part in only one association). The number of associations to which an element participates could interfere with recognition. Longer response times have indeed been observed for associations made of items participating in more associations (Sohn, Goode, Stenger, Carter, & Anderson, 2003). Increasing the number of associations could, thus, introduce interference effects in the pattern of brain activation observed for associative memory (Sohn et al., 2003), making the results more difficult to interpret.

Finally, one cannot exclude the possibility that the type of encoding influences the activation at the time of recognition. For example, the more elaborative strategy tied to associations as opposed to items might have resulted in different patterns of activation at retrieval. The effect of the type of encoding (item oriented vs. associative) on the pattern of activation during recognition of each type of material should, thus, be examined.

Beyond Differences, Common Processes to Both Item and Associative Memory

Although the focus here was on the differences that exist between item and associative memory as a function of memory stage, item memory, and associative memory

also share common underlying processes. During encoding, a common process for successful encoding has been observed with the subsequent memory effect paradigm. In this paradigm, encoding trials are separated based on the subject's subsequent memory retrieval performance. Encoding trials that lead to successful retrieval are then contrasted to encoding trials that lead to unsuccessful retrieval, revealing the brain regions implicated in successful encoding. Using this paradigm, hippocampal activation during encoding has been shown to predict the successful retrieval of both items (Brewer, Zhao, Desmond, Glover, & Gabrieli, 1998; Wagner et al., 1998) and associations (Jackson & Schacter, 2004). This suggests that beyond the modulation of hippocampal activity as a function of material as we reported in the current study, processes involved in the effective encoding of information regardless of material also modulate activity in this brain region.

Some processes recruited during retrieval are also common to both item and associative memory. Postretrieval monitoring, a process that contributes to episodic memory recognition by allowing subjects to evaluate the relevance of retrieved information in relation to the task requirements (Henson, Rugg, Shallice, & Dolan, 2000; Petrides, 2000; Henson, Rugg, et al., 1999; Henson, Shallice, & Dolan, 1999; Rugg et al., 1998; Burgess & Shallice, 1996; Rugg, Fletcher, Frith, Frackowiak, & Dolan, 1996; Shallice et al., 1994; Norman & Bobrow, 1979; Achim & Lepage, 2005), contributes to both item and associative recognition. We reported elsewhere (Achim & Lepage, 2005) that the dorsolateral prefrontal cortex shows greater activation for the type of trial associated with greater postretrieval monitoring during both item and associative recognition. During item recognition, old items are associated with greater postretrieval monitoring, and we observed greater bilateral dorsolateral prefrontal activation for this condition relative to new items. During associative recognition, rearranged associations are associated with greater postretrieval monitoring and also showed greater dorsolateral activation than intact associations.

Conclusion

In summary, our results suggest that although some structures such as the medial temporal and prefrontal cortex play a general role in memory, the pattern of activation in these regions can be modulated by the type of information (items or associations) that is being processed in memory. We observed differences in the pattern of brain activation for item and associative memory during both encoding and recognition. The hippocampus was more active during associative encoding relative to encoding of individual items, but also more active for the retrieval of items relative to associations. This pattern of hippocampal activation for item

retrieval was thought to reflect novelty detection associated with the new items. The prefrontal cortex was additionally recruited for associative encoding relative to item encoding, likely reflecting the deeper encoding strategy used for associative information. At retrieval, left prefrontal activation was observed for associative recognition, whereas item recognition predominantly recruited the right prefrontal cortex. This pattern of lateralization is consistent with the models that state that the right prefrontal cortex is involved in simple recognition judgments and that the left prefrontal cortex becomes involved for more complex recognition judgments (Lepage, 2004; Cabeza, Locantore, & Anderson, 2003). Overall, this study illustrates the importance of considering the memory stage when reporting differences between associative and item memory. The challenge remains to better understand the processes involved in encoding and retrieval of each type of information.

METHODS

Subjects

Eighteen physically healthy participants (10 men and 8 women, aged 20–50 years, mean age 29.1 years) were recruited from the community by ads in local newspapers. All were right-handed according to a sample of 12 representative items from the Edinburgh Inventory (Oldfield, 1971) (mean score = 98%, range = 86–100%). Participants were all in good health and free of any history of neurological and psychiatric disorders or substance abuse, as assessed using the nonpatient edition of the Structured Clinical Interview for DSM-IV Axis I Disorders (First, Spitzer, Gibbon, & Williams, 1998). Informed, written consent was obtained from all participants according to the institutional guidelines established by the Ethics Committee of the Montreal Neurological Hospital and Institute.

Stimuli

Stimuli consisted of 60 pairs of 2 different clipart images (referred to as pairs), 90 clipart images duplicated to have the same appearance as a pair (referred to as items), and 1 pair of 2 abstract images used as a baseline. Clipart images were obtained from a CorelDRAW picture library and depicted common objects and animals. Stimuli were divided into 2 lists (Lists I and II).

Cognitive Tasks

Participants were scanned during 4 runs, 2 encoding runs and 2 recognition runs. Half of the subjects began with encoding and recognition of List I and the other half started with List II. Stimuli were presented once every 6.5 sec. A trial consisted of the presentation of an

experimental stimulus for 2500 msec followed by a fixation cross presented for 4000 msec. When a response was required, subjects could answer at any point during the presentation of the stimulus or during the subsequent fixation cross. Presentation of the baseline stimulus during both the encoding and the recognition tasks was used to introduce some jitter in the design. This baseline stimulus was selected because it is a nonmemory condition visually similar to our stimuli that could be presented randomly with the other stimuli.

During the encoding phase, subjects were presented with a study list of 90 consecutive stimuli (30 pairs, 30 doubles, and 30 occurrences of the baseline). The order of presentation was pseudorandomized, so that each subject viewed the items in the same order. Subjects were instructed to memorize the images (both pairs and doubles) and their associations (pairs only). On each trial, subjects indicated with a mouse click whether a double or a pair was presented. No response was required for the baseline stimulus.

During the retrieval phase, subjects were presented with a list of 90 consecutive stimuli (15 old items, 15 new items, 15 intact pairs, 15 rearranged pairs, and 30 occurrences of the baseline) and were required to make memory judgments as follows: (1) when items were presented (item recognition), subjects were required to indicate with a mouse click whether it was old (studied before) or new (never studied before), and (2) when pairs were presented (associative recognition), subjects were instructed to indicate with a mouse click whether it was intact (images presented in the same pairing as in the encoding session) or rearranged (images from previously studied pairs presented in new pairings). Again, no response was required for the baseline stimulus.

To minimize set shifting and make the task easier for the subjects, item and associative recognition trials were blocked. Eight to nine recognition judgments of the same type (item or associative) were answered in a row, intermixed with presentations of the abstract stimulus. Instructions were given at the beginning of each block to inform the subjects that a switch in the type of trials had occurred, as well as to remind them of how to respond. The order of presentation was pseudorandomized so that each subject viewed the items in the same order within each list.

Subjects performed a short version of the task comprised of similar stimuli prior to the scanning session to ensure that they understood the task. They were, thus, aware of the type of recognition task that they would have to perform following the encoding session.

Scanning Procedure

Scanning was carried out at the Montreal Neurological Institute (MNI) on a whole-body 1.5-T Siemens Sonata system, using gradient-echo EPI sequences. A vacuum cushion stabilized the subject's head. Stimuli were gen-

erated by a Pentium-class PC laptop computer and projected via an LCD projector and mirror system. A mouse connected to the computer recorded the subject's responses. Each scanning session began with a high-resolution T1-weighted 3-D volume acquisition for anatomical localization (voxel size $1 \times 1 \times 1$ mm). This was followed by the acquisition of the functional images of the brain. T2*-weighted images were acquired with blood oxygenation level-dependent contrast (TR = 3000 msec, TE = 50 msec) and covered the entire brain (25 slices, in-plane resolution: 2×2 mm, thickness: 5 mm). Functional scans were acquired parallel to the anterior–posterior commissural plane. Each functional run consisted of 230 scans. The first scan of each run triggered the start of the cognitive task.

Functional Magnetic Resonance Imaging Data Processing and Statistical Analyses

The T2* images were first realigned to the fifth image in their respective run and spatially smoothed with a 6-mm full width half maximum (FWHM) isotropic Gaussian kernel. fMRI images were then analyzed with fmristat (Worsley et al., 2002) using an event-related procedure and the default parameters. The statistical analysis of the fMRI data was based on a linear model with correlated errors. Three event types were modeled for the encoding task (encoding of items, encoding of pairs, and abstract stimulus presentation) and 6 event types were modeled for the recognition task (correct recognition of old doubles, correct recognition of new doubles, correct recognition of intact pairs, correct recognition of rearranged pairs, baseline stimulus, and errors irrespective of the stimulus type). The errors were, thus, excluded from the contrasts of interest. The design matrix of the linear model was convolved with a hemodynamic response function modeled as a difference of 2 gamma functions timed to coincide with the acquisition of each slice. Low frequency drifts were removed by including polynomial covariates, up to degree 3, in the design matrix. The correlation structure was modeled as an autoregressive process of degree 1. At each voxel, the autocorrelation parameter was estimated from the least squares residuals using the Yule–Walker equations. The autocorrelation parameter was first regularized by spatial smoothing with a 15-mm FWHM Gaussian filter, and then used to whiten the data and the design matrix. The linear model was then reestimated using least squares on the whitened data to produce estimates of effects and their standard errors. In a second step, the 2 runs of encoding and the 2 runs of recognition were combined using the effects and standard deviations images taken from the previous analysis. This was fitted using restricted maximum likelihood estimates implemented by the maximization (EM) algorithm. A random-effects analysis was performed by first estimating the ratio of the random-effects variance to the fixed effects

variance, then regularizing this ratio by spatial smoothing with a 15-mm FWHM Gaussian filter. The variance of the effect was then estimated by the smoothed ratio multiplied by the fixed effects variance to achieve higher degrees of freedom (Worsley et al., 2002). The resulting effect and standard deviation images were then normalized to standard space using the MNI template (Cocosco, Kollokian, Kwan, & Evans, 1997) as a reference. Group analyses were then performed using the same procedure that was used to combine the runs, but this time with a smoothing of the variance ratio of 8 mm.

Contrasts were performed between item and associative encoding trials and between item and associative recognition trials. Conjunction analyses were then performed to identify brain regions activated during both encoding and retrieval of each type of information relative to the other type. Statistical analyses were performed separately at each voxel. For our 2 regions of interest, the medial temporal lobe and the prefrontal cortex, the threshold was set to $p < .01$, uncorrected for multiple comparisons, $t(137) = 2.35$ for the contrasts and $t(137) = 1.29$ for the combined probability in the conjunctions analyses, and $p < .001$, uncorrected for multiple comparisons, $t(137) = 3.15$ for the contrasts and 1.87 for the conjunctions, respectively, according to the size of the respective regions. We considered as part of the medial temporal cortex the amygdala, hippocampus, and parahippocampal gyrus, and as part of the prefrontal cortex, the fronto-polar cortex (Brodmann's area [BA] 11/47), frontal pole (BA 10), ventrolateral prefrontal cortex (BA 44/45), and dorsolateral prefrontal cortex (BA 46/9/8). All other areas of the brain were examined with a threshold of $p < .05$, corrected for multiple comparisons, $t(137) = 4.14$ for the contrasts and 2.58 for the conjunctions.

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The data reported in this experiment have been deposited with The fMRI Data Center archive (www.fmridc.org). The accession number is 2-2005-118BJ.

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